

Target-Guided 3D Reconstruction Based on Tello Drone



Task Modeling

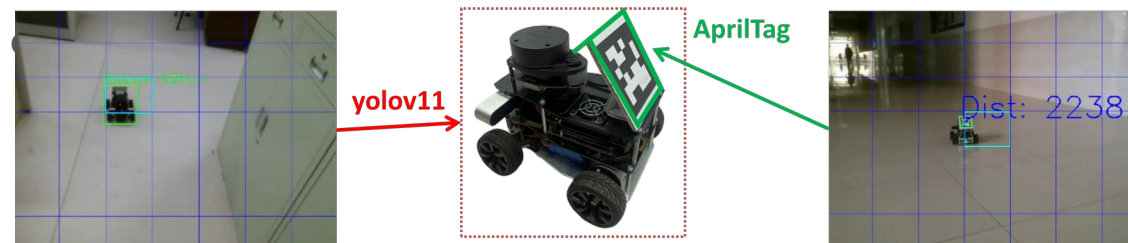
1. Drone follows target vehicle, maintaining visual contact through flight control.
2. Dynamic obstacle avoidance with algorithms ensuring post-avoidance tracking.
3. RGB-based 3D reconstruction, excluding moving vehicle from static scene.

Key Challenges

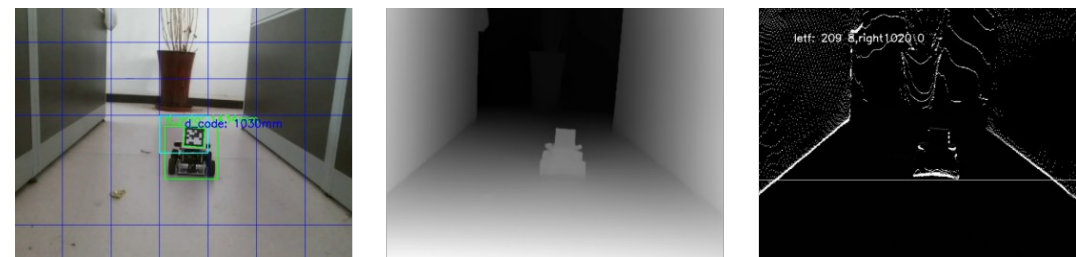
1. Multi-view vehicle tracking & distance estimation
2. RGB-only depth estimation
3. Obstacle detection & control
4. LiDAR/sonar-free 3D reconstruction
5. Dynamic object filtering

Technical Analysis

1. YOLO + AprilTag for tracking and distance



2. DepthAnything for depth estimation, Sobel operator for edge-based obstacle avoidance while tracking vehicle.



3. Mast3r-SLAM for 3D reconstruction, YOLO filters moving objects.





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PART 02

AprilTag-Based Tracking with YOLOv11

Benefits of YOLOv11 & AprilTag

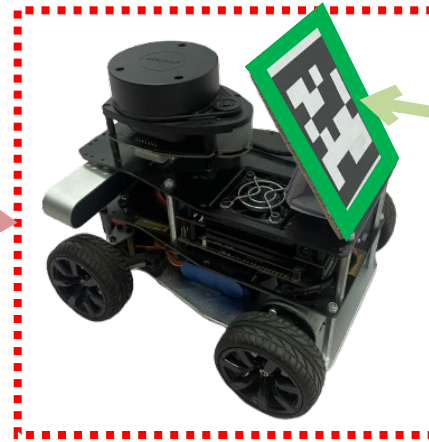


Yolo Features

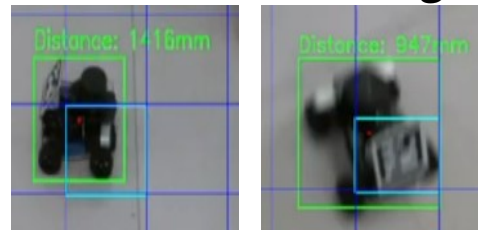
Adv: Various angles

Disadv: distance estimation is not accurate.

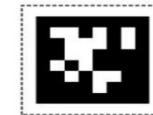
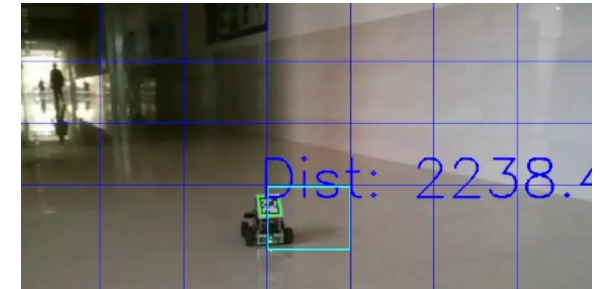
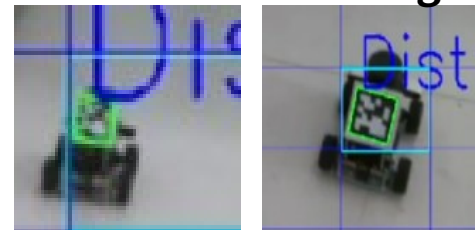
Dual Detection Advantage:
YOLO flexibility + AprilTag accuracy



YOLO extreme condition testing



Apriltag extreme condition testing



Apriltag Features

Adv: Precise distance measurement from the front

Disadv: Less robust to angle changes

Target guide tracking

01

Target information processing

1. Grid overlay
2. YOLO detection
3. Rough distance measurement
4. Precise distance measurement

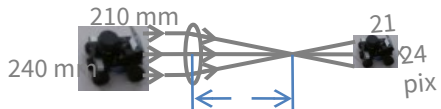
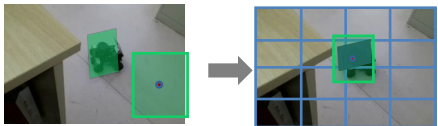
Case 1. with AprilTag
Case 2. without AprilTag

Flight controller matrix

Overlay a 7×7 grid



With grid cell containing target

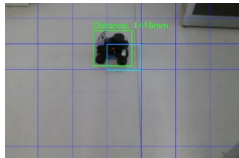


$$yolo_distance = \frac{f \times \sqrt{area_real}}{\sqrt{area_pix}}$$

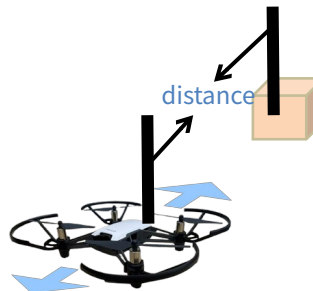
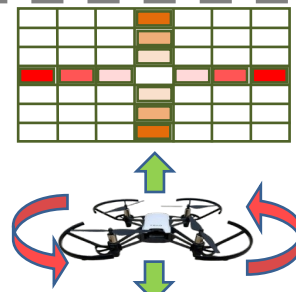
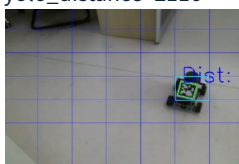


AprilTag_distance
Camera_post
APrilTag_location

Case1: without AprilTag
Apriltag_distance=null
yolo_distance=1416



Case2: with AprilTag
Apriltag_distance=2100
yolo_distance=2116



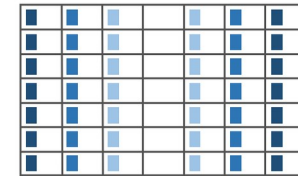
02

Flight controller matrix

Velocity decreases from edges to center
(30cm/s → 0, decreasing 10cm/s per step).

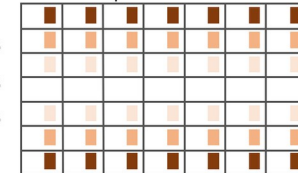
Left-Right FCM

- speed_lr=30cm/s
- speed_lr=20cm/s
- speed_lr=10cm/s

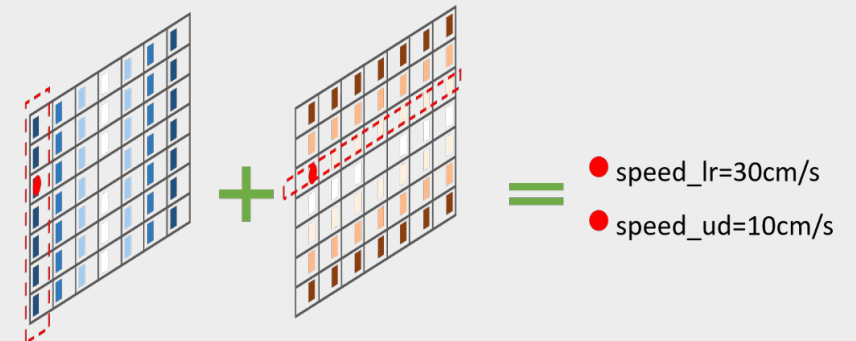


Up-Down FCM

- speed_ud=30cm/s
- speed_ud=20cm/s
- speed_ud=10cm/s



e.g. Located at row 3, column 1, denoted as A(1,3)
At this point, speed_lr= 30 cm/s, speed_ud= 10 cm/s.



PART 03

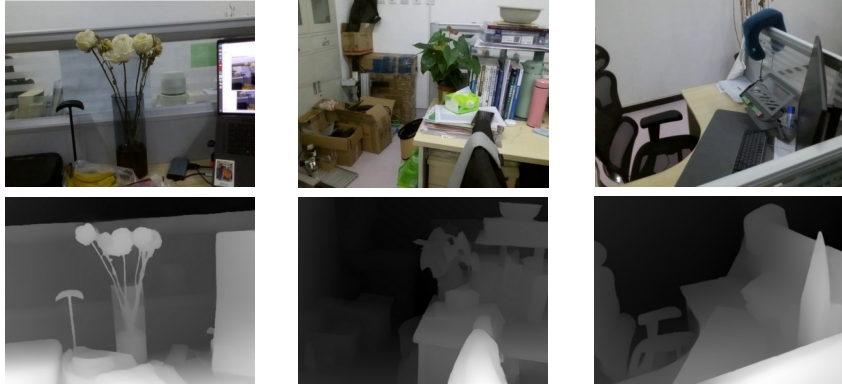
Dynamic Obstacle Avoidance



DepthAnything & Sobel operator

01 Depth Map Creation

Depth Anything serves as a substitute for depth sensing hardware. Depth values are normalized to the range 0-255

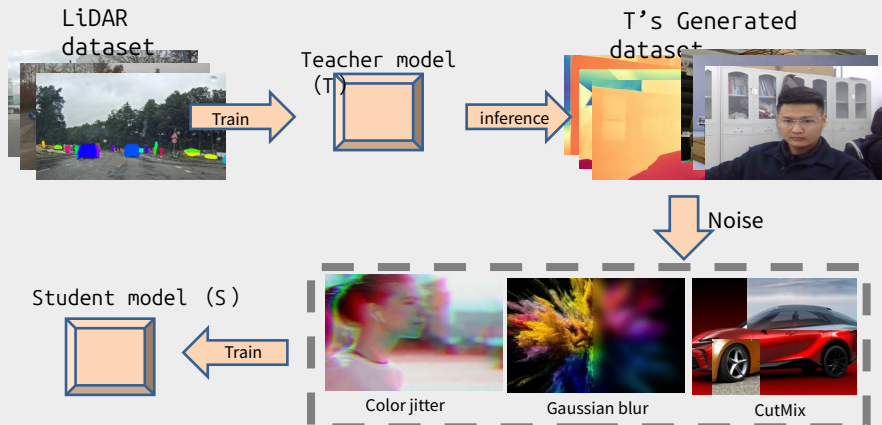


Test case 1

Test case 2

Test case 3

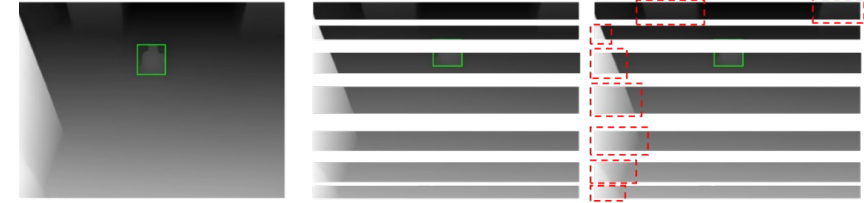
Employs knowledge distillation: a **teacher model** trained on large labeled datasets guides a **smaller student model** to achieve the same task with minimal accuracy loss.



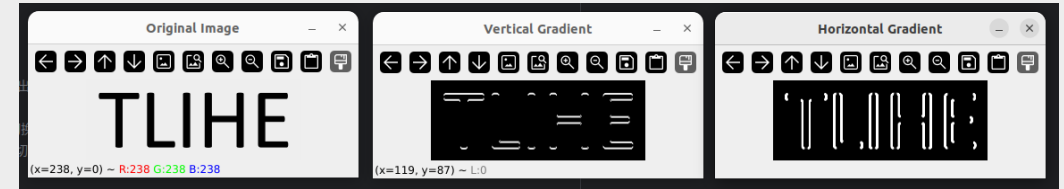
02

Sobel operator obstacle detection

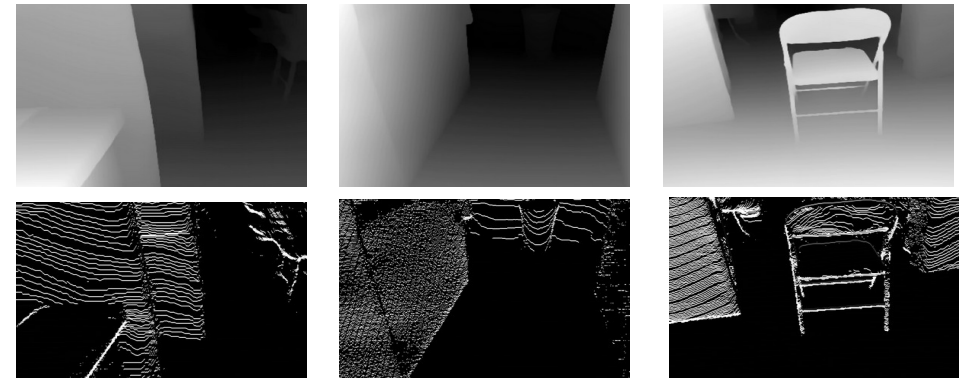
“Ground” depth maps show **linear variation** with distance (left). Horizontal slicing reveals that most areas have minimal gradient change, while **significant changes indicate obstacles** (right).



Sobel is a **gradient-based** edge detector using X and Y directional gradients. Here, only the horizontal filter is applied.



comparison between the depth map and obstacle detection map.



Test case 1

Test case 2

Test case 3

PART 04

3D Reconstruction System



MASt3R-SLAM

Principle & improvements

Three stages of MASt3R-SLAM

Initialization stage

Creates the first keyframe and initial map by processing the first frame through MASt3R to generate pointmap and encoded features.

Tracking stage

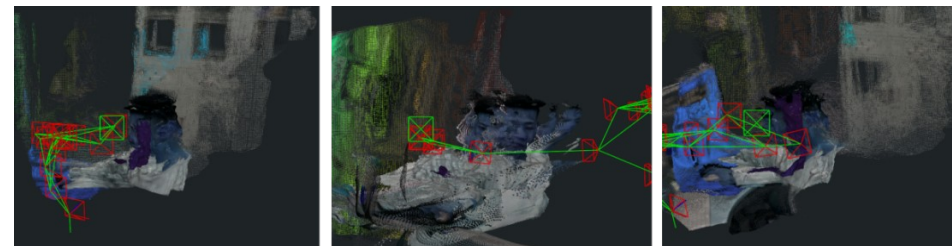
Estimates current frame pose through efficient pointmap matching with recent keyframes and performs local map fusion.

Relocalization stage

Recovers tracking by matching current frame against the global map database when tracking fails or error accumulates.

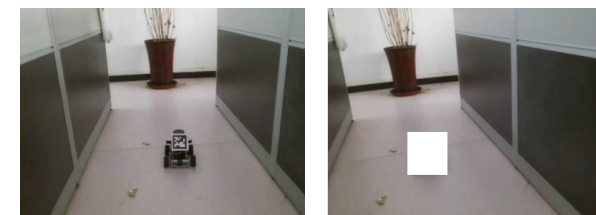
Challenge: Dynamic objects impact static environment

Moving objects interfere with vehicle-guided drone mapping by causing feature matching errors that deform static point clouds.



Solution: Vehicle Removal

Expand YOLO detection region by 3 pixels outward and apply a transparent mask over the vehicle area.



Implementation: Add Alpha Channel

1. Use alpha channel to convert input image to RGBA format
2. Adjust initialization to skip tracking phase for masked regions
3. Set confidence to 0 in masked areas

original_confidence

0